Does the LD panel matter for PGS?

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github.com/fboehm/linkage-disequilibrium-panels

Background

Polygenic scores (PGS) trained on GWAS summary statistics need an external linkage-disequilibrium (LD) reference panel. In practice that panel is often **ancestry-mismatched** or **undersized** relative to the GWAS. How much PGS accuracy do we lose?

Simulation design

Genotypes: HAPNEST, six superpopulations (AFR, AMR, CSA, EAS, EUR, MID) on 22 autosomes.

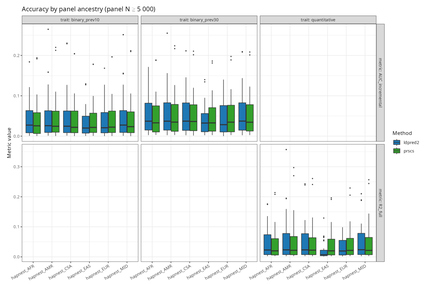
GWAS: $N_{\text{gwas}} = 100{,}000$; held-out target $N = 5{,}000$.

Phenotypes: $h^2 \in \{0.05, 0.10, 0.20, 0.40\}$, $p_{\text{causal}} \in \{0.001, 0.01, 0.05, 0.20\}$, infinitesimal vs. spike-and-slab effects, three trait types.

LD panels: $N \in \{500;\ 5{,}000;\ 50{,}000\} \times$ six panel ancestries.

PGS methods: LDpred2 & PRS-CS.

Methods compared

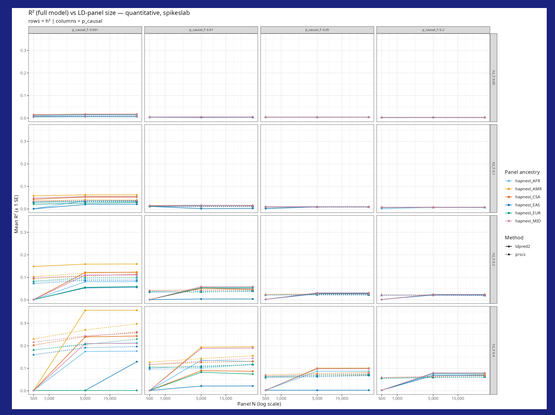


LDpred2 and PRS-CS deliver near-identical accuracy across ancestries and traits.

Panel ancestry and size only matter when the trait is **sparse** and **highly heritable.**

In infinitesimal regimes,

any panel will do.



R^2 vs. LD-panel size, spike-and-slab effects. Rows: heritability h^2. Cols: causal-SNP fraction p_{causal}. Curves separate (matched AMR best, mismatched EAS worst) **only** in the lower-left tiles — sparse *and* heritable.

When the panel matters

Sparse, high-h^2 traits ($p_{\text{causal}} \leq 0.01$, $h^2 \geq 0.2$): matched-ancestry AMR panels reach $\sim 2\times$ the R^2 of the worst mismatched panel (EAS).

Panel size drives most of its gain between $N = 500$ and $N = 5{,}000$; the jump to $N = 50{,}000$ adds little.

When it doesn't

Infinitesimal traits (gaussian effects): all six panel ancestries cluster within noise; panel size has negligible effect.

Low-h^2 traits ($h^2 \leq 0.10$): PGS accuracy is floor-limited; panel choice is a second-order concern.

Implications

For most polygenic traits the LD-panel choice is a *minor* contributor to PGS accuracy. Investment in matched-ancestry panels pays off **primarily for monogenic-leaning, high-heritability traits**.

Code & data

Snakemake pipeline; HAPNEST genotypes; LDpred2 and PRS-CS posteriors. All metrics in `results/summary/all_metrics.tsv`.

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